INDEPENDENT SOFTWARE PROJECT / CASE STUDY

UMA-BESTDERBY

A client-side lineup optimizer for Umamusume Team Trials

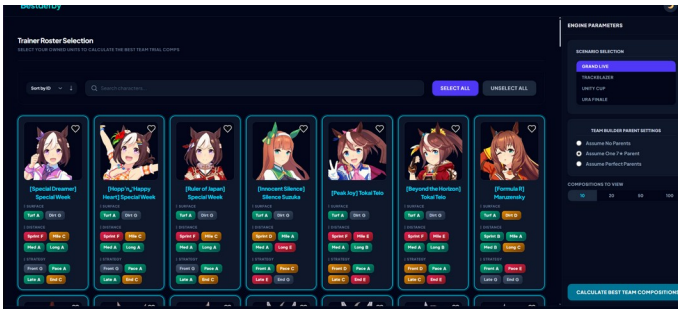
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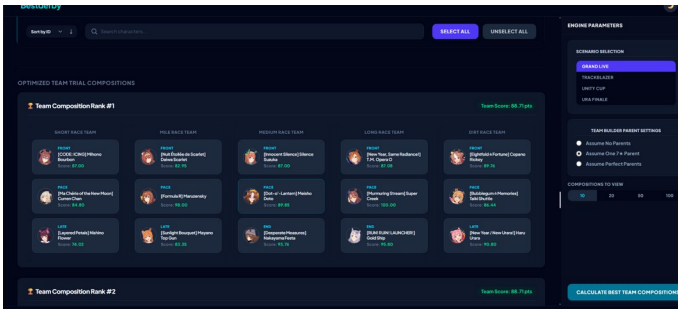
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AT A GLANCE Players select the characters they own, choose a training scenario and an inheritance setting, then receive ranked five-team compositions based on estimated scoring potential.

THE PRODUCT From roster and settings to ranked team recommendations



01 SELECT ROSTER + PARAMETERS



02 REVIEW RANKED TEAM COMPOSITIONS

WHY IT EXISTS

THE CHALLENGE

Team Trials requires five squads of three characters with no repeats. Every pick involves track and running-style aptitudes, skills with conditional activations, inheritance options, and penalties for conflicting running styles.

THE SOLUTION

The application evaluates a player's available roster against precomputed character scores and lineup constraints, then displays ranked compositions across Sprint, Mile, Medium, Long and Dirt race categories.

HOW IT WORKS

SCORING + OPTIMIZATION

Character scores are indexed by scenario → parent setting → track → running style. Unsuitable aptitudes receive zero scores unless the selected inheritance setting makes them eligible (base aptitudes, a 7-star parent, or perfect parents). The React/TypeScript algorithm runs in the browser, prevents character reuse across squads, and disallows duplicate running styles within a squad.

DATA + APPLICATION ARCHITECTURE

A nightly rclone job copies the local game master.db from a gaming PC to Backblaze B2. A separate script parses game information and updates MongoDB with character details and scores. The app uses React, TypeScript, Node.js and Express; MongoDB supplies the underlying data.

TOUGHEST ENGINEERING DECISION

Turning conditional game mechanics into usable scores. Skill value depends on whether it actually activates, not merely its theoretical effect. I consulted experienced players to develop more defensible evaluations. The optimizer uses these estimates to recommend lineups designed to maximize scoring potential, even when individual race results vary.

TECHNOLOGIES React · TypeScript · Node.js / Express · MongoDB · Backblaze B2 · rclone · Tailwind CSS · Render

Built independently · Source repository remains private · View live application